



Accessible ACD Screening for Chronic Eczema Patients

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Need Statement

A way to increase accessibility of contact dermatitis screening for people with chronic eczema in order to facilitate timely allergen identification and management



[Image Source](#)

The Problem

Traditional ACD screening is inaccessible to approximately 98% of the US population

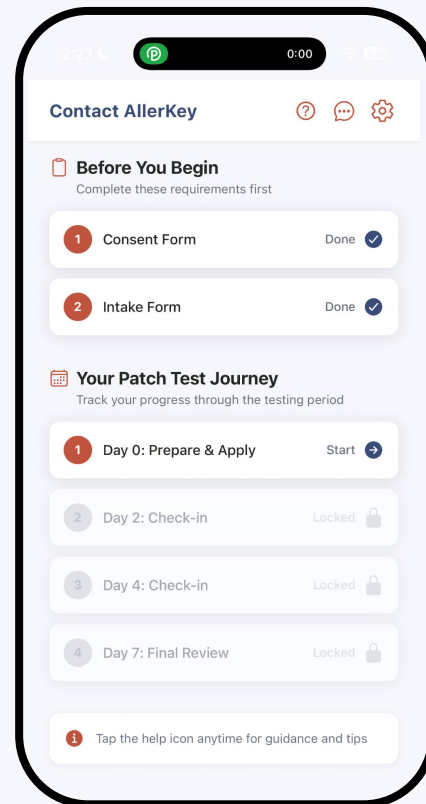
The Impact

~20% of the US population suffers from preventable flare-ups due to unknown triggers

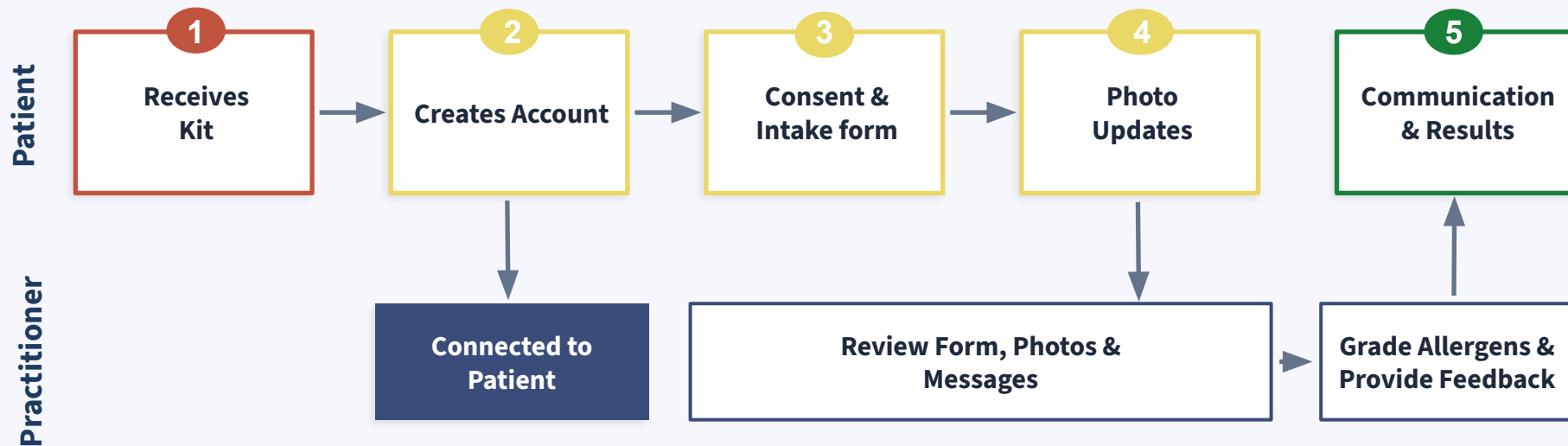
The Opportunity

Innovate patch testing to make identifying allergens accessible; completely ending experience of symptoms for patients

Components of AllerKey



User Experience Flow



Onboarding Flow - Onboarding Flow & Entry Points

App Flow

- Welcome screen — logo + Sign In / Create Account
- What to Expect — Day 0, 2, 4, 7 timeline
- Reassurance — 4 trust-building cards
- → Login or Register

Auth Flow

- New User: Select the User type (Patient or Practitioner)
 - Name
 - Email
 - Password
- Returning User:
 - Log in

Entry Points

- Direct app launch
- Push notification (Day 2/4/7 reminders)
- Web: 2026-skin.vercel.app (practitioner)
- Expo Router

Our Journey

Scratch That PRD — Vision

- Patient-driven home patch test
- Login with DUO authentication
- Informed consent form
- Medical intake questionnaire
- Day 0 photo capture (patch application)
- Patient dashboard (status tracker)
- Step-by-step tutorial guide



Contact AllerKey — Shipped

- Full 7-day protocol: Day 0 → 2 → 4 → 7
- 2-layer AI photo QA (local checks + GPT-4V)
- Practitioner dashboard + 36-allergen grading
- FHIR-compliant backend (Medplum EHR)
- Smart push notification reminders
- Cross-platform: iOS · Android · Web
- Dual-persona: patient ↔ practitioner

Branding

“Scratch That” — internal prototype name

➡ **Contact AllerKey — patient-facing clinical brand**

Backend Architecture

Medplum

➡ **100% Medplum FHIR — unified identity & data**

Product Scope

Patient-only guided test app

➡ **Dual-persona: Patient app ↔ Practitioner dashboard**

Photo Assessment

Basic camera capture at Day 0 only

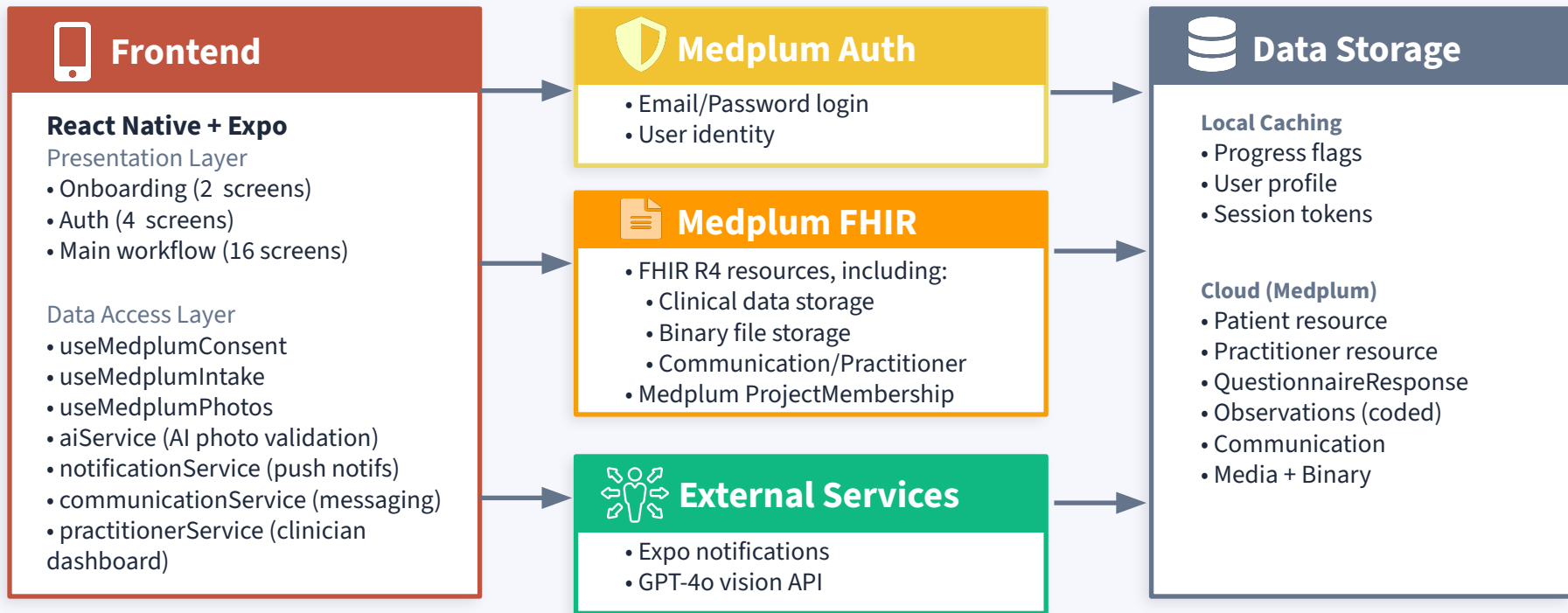
➡ **4-stage AI quality gate across 7-day protocol (GPT-4V)**

Tools Used

Stanford SpeziVibe	Claude Code	Cursor	Expo Go
			
• Free and open-source • Has the components that cover the most common functionality needed for a modern, scalable digital health solution • Integrates directly with electronic health record systems Spezi through HL7 FHIR	• Credits given to us by Course Staff • Lives in your terminal or IDE and understands your entire codebase • Can work directly with git, i.e. tag changes, write commit messages, create branches, and even open pull requests	• A vibecoding tool that works directly with git to generate code, identify changes, write commit messages, create branches, and even open pull requests • Lives in your terminal or IDE and understands your entire codebase	• Real-time reloading allows us to instantly see the impact of a fix or implementation of a new feature • Supports various platforms (web, IOS, android) • Integrates third party services well and provides detailed error logs

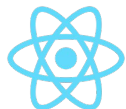
System Architecture

Three-Layer Architecture with Cloud Backend



Tech Stack

Frontend



React Native

Authentication



Backend



Storage



Clinical Standards



FHIR Resources

Patient, Practitioner
QuestionnaireResponse
Observation
Communication
Media / Binary

Cross-platform, healthcare-compliant

Using Medplum

- Helpful documentation across all aspects, from authentication to data storage
- Binary storage for medical images ensured that we didn't need S3 or Firebase storage separately
- Unified user identifiers + IAM as opposed to linking with Firebase auth
- Admin dashboard for debugging and data inspection

What's Next

- Patent and FDA approval is pending
- Applying to Catalyst
- Applying for Coulter grant
- Small-batch production
- Patient trials